

BAEK SEONG-EUN

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Education

Pohang University of Science and Technology (POSTECH) 2024.08-present

Master student of Artificial intelligence

Korea university 2019.03-2024.08

Bachelors of Biomedical Engineering & Artificial intelligence , 4.0/4.5 GPA

- Major in Biomedical Engineering
- Minor in Artificial intelligence

Research Interest

- Multi-modal, Video understanding, Bayesian Optimization with LLM

Experience (Graduate.)

Project with Samsung Electronics AI Center (SAIT) 2025.03-present

- Multi-modal expansion by taming cross-modal representation
- Visualization Optimization
- LoRA Hyperparameter Optimization

Project with Korea Research Institute for Defense Technology 2024.09-2025.10

- Anomaly Detection via Contextual Modeling of Multi-Target

Teaching Assistant in Samsung SW · AI academy For Youth (SSAFY) 2024.10-2024.12

- Large Language Model (LLM) and Foundation Generative Model
- Retrieval-Augmented Generation (LangChain)

Experience (Undergraduate.)

MLV lab 2024.01.02-2024.03.31

- In-Context Retrieval Augmented Language Modeling in VQA
- Subsequent study of Flipped-VQA

Brein lab 2023.09.04-2023.12.21

- Medical foundation model (BERT) training using NHANES data
- Medical foundation model (BERT) training using UKBiobank data

NeuroXT 2023.01.02-2023.08.31

- Predicting Alzheimer's Disease Progression in Neuro-imaging using Severance medical data
- Develop brain image visualization tool

Data science society in Korea University (KUBIG) 2023.08-2024.08

- Reproducing causal inference language model (CausaLM)
- Open-source LLM(ex. LLaMA) fine-tuning
- Generate synthetic data using Dalle2 model and train classification model
- OpenFlamingo Term project

Samsung Securities IT Society Linkage Program 2023.09-10

- Develop a question answering model for a given document
- Text parsing, graph detection, model development and fine-tuning

Military Service 2020.07.13-2022.01.13

- Drill Sergeant of the Army Infantry School in Hampyeong-gun

Term project

- Rating prediction system using Naver Webtoon's best comments
- Road traffic forecast in Jeju
- Sentiment Analysis using binary classification dataset

Publications

[P7] Lee Jung-Mok, **Baek Seong-Eun**, Kim Sung-Bin, Tae-Hyun Oh, “Learning to Visualize: Visual Presentation Optimization for Time-Series Analysis” *preprint*, 2026 (Under-Review in ECCV).

[P6] Yejin Yeo, **Baek Seong-Eun**, Shin Dong-Yeon, Lee Jung-Mok, Tae-Hyun Oh, “TypoErase: Training-Free Image Level Defense against Typographic Attacks” *Workshop on Image Processing and Image Understanding (IPIU)*, 2026 (Under-Review in ACL).

[P5] **Baek Seong-Eun**, Lee Jung-Mok, Kim Sung-Bin, Tae-Hyun Oh, “A Language-Guided Bayesian Optimization for Efficient LoRA Hyperparameter Search” *ICML*, 2026.

[P4] Kyeongseon Kim, **Baek Seong-Eun**, Lee Jung-Mok, Tae-Hyun Oh, “Training-free Multimodal Embedding for Structure-Aware Retrieval of Scalable Vector Graphics and Images,” *WACV*, 2026.

[P3] **Baek Seong-Eun**, Nam Hyeon-Woo, Lee Jung-Mok, Tae-Hyun Oh, “Do MLLMs Understand Arrows? Visual Motion Prompting and Motion-Aware Reasoning Make Motion Easier to Understand” *ICCV workshop*, 2025. (Under-Review in ICML).

[P2] **Baek Seong-Eun**, Nam Hyeon-Woo, Tae-Hyun Oh, “Can Red Arrow Enhance Fine-Grained Video Understanding?” *Workshop on Image Processing and Image Understanding (IPIU)*, 2025.

[P1] **Baek Seong-Eun**, Moon Ye-Bin, Nam Hyeon-Woo, Tae-Hyun Oh, “교통사고 예방을 위한 검색 증강 생성 (RAG) 시스템” *Korea Robotics Society Annual Conference (KRoC)*, 2025.

Patents

시각-거대언어모델 대상 시각화 구성 파라미터의 베이지안 최적화 기반 탐색 기술 On-going

Awards

Samsung Securities IT Contest Excellence Award

Skills

Programming: Python, MATLAB
English: TOEIC 860